Sample Resume – Second Year M.S. Chemical Engineering

ROHIT T. MOHAMMAD

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EDUCATION

College Park, MD

UNIVERSITY OF MARYLAND — GPA: 3.7 — Expected May 2024
M.E. Chemical Engineering

UNIVERSITY OF VIRGINIA — Charlottesville, VA
B.S. Chemical Engineering, Business Minor — GPA: 3.8 — May 2018

- Magna cum Laude

SKILLS AND CERTIFICATIONS

Software: MATLAB, MathCAD, ASPEN, ANSYS, Microsoft Office (Word, Excel, Access and PowerPoint)
Laboratory: Gas Chromatography, organic synthesis & purification, HPLC, atomic absorption

RELEVANT EXPERIENCE

ABC DEVELOPMENT COMPANY
Energy Sector Analyst — July 2018 - August 2022
Verification of Enhanced Oil Recovery Audits — Toronto, Canada

- Assessed whether the quantity of offsets generated was characterized accurately by evaluating calculation methodologies, re-performing direct and indirect emissions calculations, and analyzing P&IDs of the injection and production facility.
- Identified compressor seals and CO2 dissolved in stored crude as the missing emission sources that accounted for 22% of the total emissions from the site.

Greenhouse Gas Emission and Sinks Inventory, Oil and Gas Sector — Washington, DC

- Established the uncertainties associated with each emission source using @RISK, a Monte Carlo simulator capable of performing risk analysis in Excel spreadsheets.
- Improved the U.S. Inventory by researching and quantifying the emissions reductions from technologies and processes reported by Natural Gas STAR Partners to enhance the assumptions and the resulting emissions estimates.
- Recommended statistical methods to validate the emissions profile of a facility and estimated expected ranges for data elements collected through reporting.

Measurement Study for Indian Natural Gas Industry — Bhopal, India

- Designed Excel-based tools to aggregate emissions data into a comprehensive emissions inventory and perform economic analysis of mitigation options for the major emission sources; analysis revealed savings of $3.75 million.
- Shared results of the measurement study and mitigation options to EPA client in a technical presentation.

Energy Performance Benchmarking and Conservation Potential — Washington, DC

- Established a baseline energy consumption profile for equipment used in transmission processing of oil and gas by researching Title V permits, vendor documents, and by using engineering calculations.
- Developed an estimate of potential energy savings by creating a database of conservation methods and using a proprietary calculation platform.

Climate Business Opportunities Washington, DC

- Identified key investment opportunities to target as climate change and sustainability become increasingly important in developing countries.
- Inventoried and projected the growth of emissions from the oil and gas industry in developing countries based on key market metrics.
- Estimated reduction potential for emissions by evaluating the implementation of mitigation technologies, including the potential for miniaturized gas-to-liquid technologies.
- Conducted a rigorous search of potential companies within the target regions that met the client's stringent investment criteria.

ABC ENERGY PARTNERS Bethesda, MD
Project Coordinator June 2014 - June 2016

- Delivered technical presentations on emission mitigation technologies and practices and provided logistical support for numerous technology transfer workshops.
- Researched and organized data on process units at refineries
- Calculated the equivalent distillation capacity of each refinery in the U.S. that took into consideration the complexity of process units at each refinery.

RESEARCH EXPERIENCE

PETROLEUM INSTITUTE Abu Dhabi, U.A.E
Research Assistant June - August 2013

- Established a theoretical formalism linking thermal and visco-elastic properties of crude oil used in reservoir simulations and enhanced oil recovery methods.
- Published results of research:
  - Ayaz, A.; **Mohammad, R.T**., Temperature dependent thermodynamic and thermo-elastic properties of crude oil. Journal of Engineering Topics 2012, vol. 5, pp 123-145.
  - **Mohammad, R.T**., Perez, P., Visco-elastic and dielectric relaxation studies of crude oil. Petroleum Science and Technology 2013, vol.21, pp 234-344.

AFFILIATIONS

American Institute of Chemical Engineers, Events Committee Member September 2012 - Present
Association of Energy Engineers, Member May 2014 - Present
Tau Beta Pi Honor Society, Member September 2013 - May 2014